

Topic 4

Interference and Diffraction

(Halliday/Resnick/Walker Ch. 35, 36)

Content

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- Young's Interference Experiment
- Intensity of Interference Pattern
- Thin Film Interference
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- Introduction to Diffraction
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- Diffraction Grating
- Diffraction Resolution Limit

Interference of Light/EM Waves



soap film



soap bubbles



CD/DVD



Analog TV



butterfly



peacock



beetle

Structural color: peacock tail has a brown pigment but their microscopic structure makes it reflect blue and green color due to interference

What is Interference?

- **Interference:** a physical phenomenon in which two or more waves superimpose to form a resultant wave. It usually refers to the interaction of waves that are **coherent or correlated** (i.e., relative phase difference is not random in time) with each other.
- The resultant wave is obtained by the **superposition principle**. Its amplitude depends on the phase difference between the interacting waves,
 - **In-phase (constructive)** interference → amplitude doubling
 - **Out-of-phase (destructive)** interference → amplitude cancelling
- **Optical interference** is the interference between light waves and is applied in many branches of science and engineering.

For example:

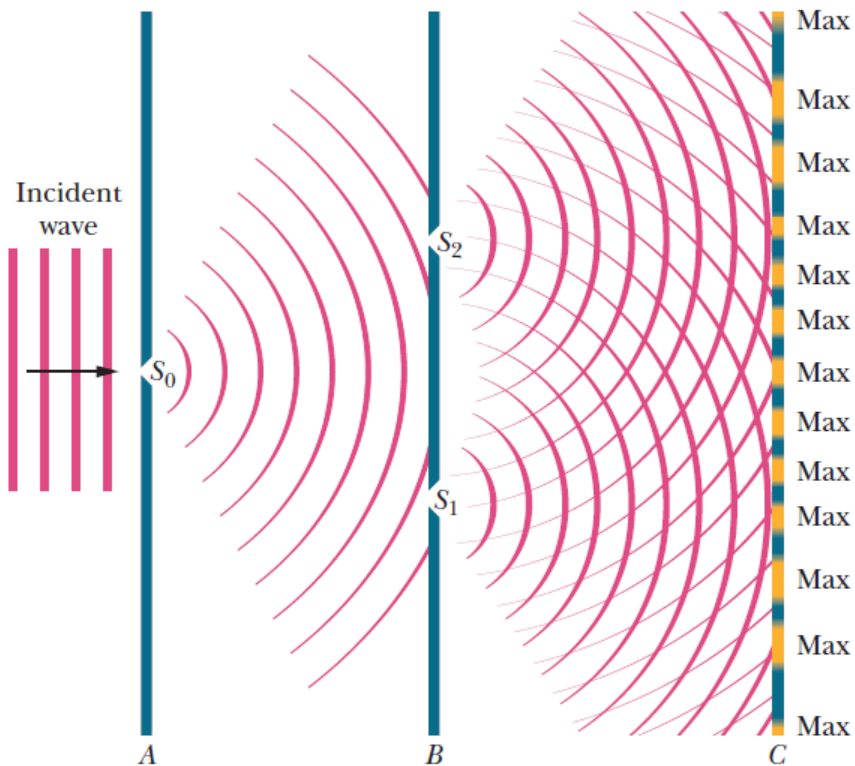


The blue color on the top surface of a *Morpho* butterfly wing is due to optical interference. It shifts in color as you change your viewing angle.

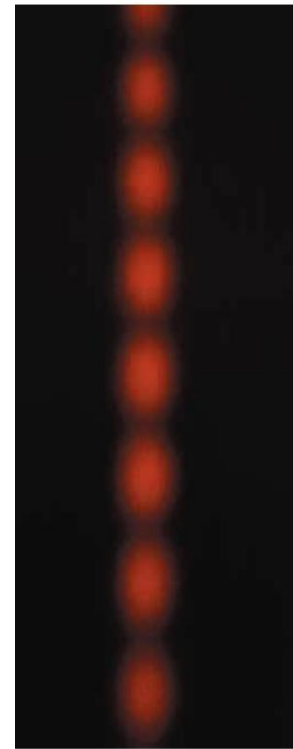
Young's Interference Experiment

Young's interference experiment (double slit interference experiment): ~1801

Significance: proved that light is a wave, contradicting Isaac Newton's view



Formation of wave interference pattern

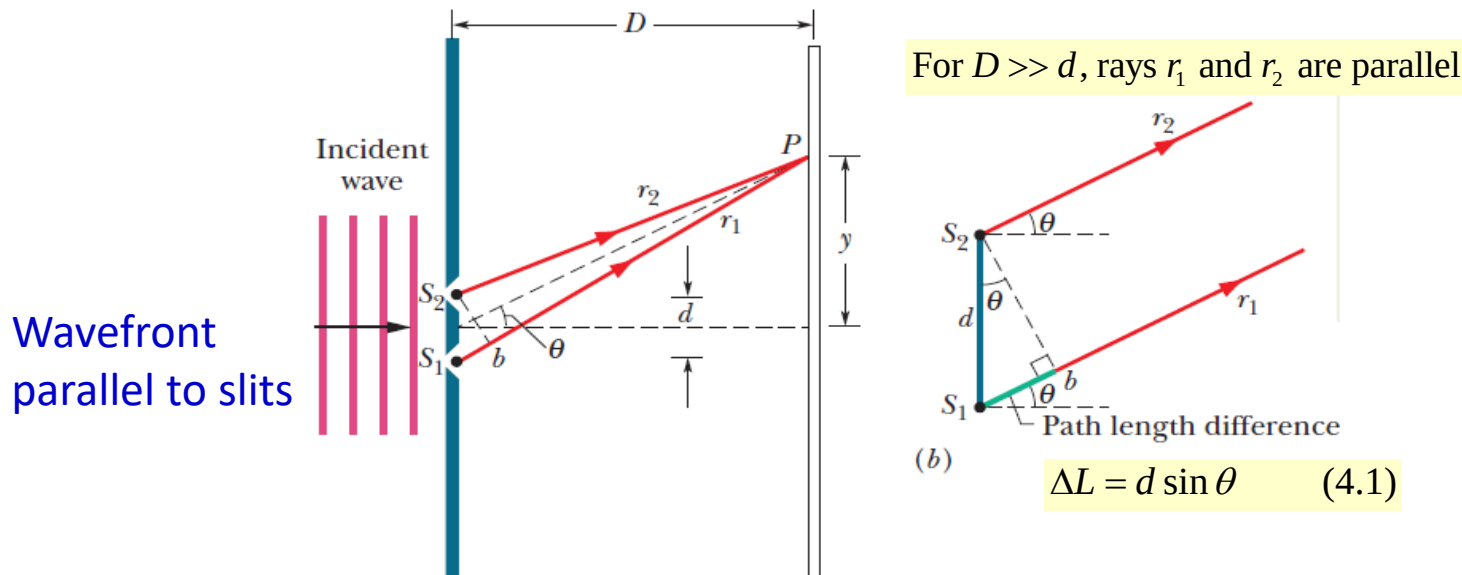


interference fringes



Thomas Young
(1773–1829)
English scientist
(Also Young's modulus)

Optical Path Difference and Interference Pattern



- **Bright fringes (constructive interference, maxima):** ΔL must be either zero or an integer number of wavelengths. Maxima appear at angles θ satisfying $d \sin \theta = m\lambda$, $m = 0, 1, 2, \dots$ (4.2)

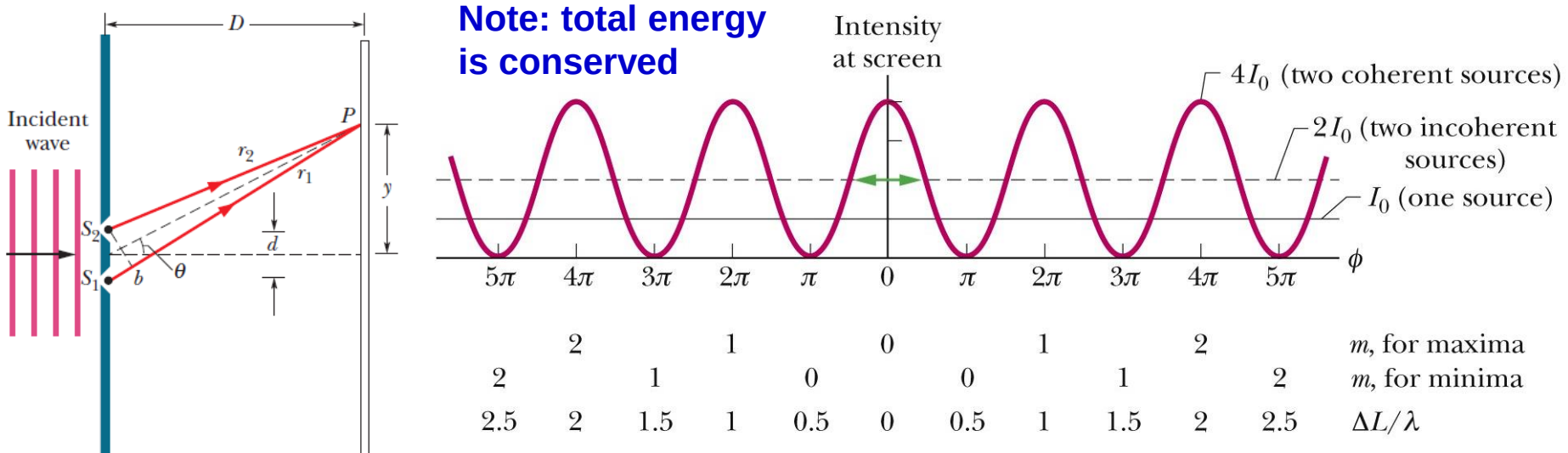
- **Dark fringes (destructive interference, minima):** ΔL must be an odd multiple of half a wavelength. Minima appear at angles θ satisfying $d \sin \theta = \left(m + \frac{1}{2}\right)\lambda$, $m = 0, 1, 2, \dots$ (4.3)

- **Spacing between adjacent bright fringes:** when θ is small, $\sin \theta \approx \theta \approx \tan \theta = \frac{y}{D}$ (4.4)

$$y_{m+1} - y_m = D \sin \theta_{m+1} - D \sin \theta_m = D \frac{(m+1)\lambda}{d} - D \frac{m\lambda}{d} = \frac{D\lambda}{d} \quad (4.5)$$

$\therefore$ Spacing between neighboring bright fringes remains unchanged when θ and d are small.

Intensity of Interference Pattern



Consider electric field components of the light waves at point P on the screen:

$$E_1 = E_0 \sin \omega t \quad (4.6)$$

$$E_2 = E_0 \sin(\omega t + \phi) \quad (4.7)$$

constant phase difference $\rightarrow$ coherent

It can be shown (proof to follow) that the intensity (power/area) at point P is

$$I = 4I_0 \cos^2\left(\frac{\phi}{2}\right) \quad (4.8) \quad \text{where } \phi = \frac{2\pi d}{\lambda} \sin \theta \quad (4.9)$$

Intensity of Interference Pattern

Proof:

Recall $\sin A + \sin B = 2 \cos\left(\frac{A-B}{2}\right) \sin\left(\frac{A+B}{2}\right)$

Hence, the electric field at point P is

$$E = E_1 + E_2 = E_0 \sin \omega t + E_0 \sin(\omega t + \phi) = 2E_0 \cos\left(\frac{\phi}{2}\right) \sin\left(\omega t + \frac{\phi}{2}\right) \quad (4.10)$$

constant amplitude

Thus, the electric field amplitude is changed by a factor of $2 \cos\left(\frac{\phi}{2}\right)$.

Within the same medium, Intensity $\propto$ (field amplitude)², hence changed by $4 \cos^2\left(\frac{\phi}{2}\right)$.

$$I = 4I_0 \cos^2\left(\frac{\phi}{2}\right) \quad (4.8)$$

with phase difference $\phi = \frac{2\pi}{\lambda} \cdot \underbrace{d \sin \theta}$

$$\rightarrow \phi = \frac{2\pi d}{\lambda} \sin \theta \quad (4.9)$$

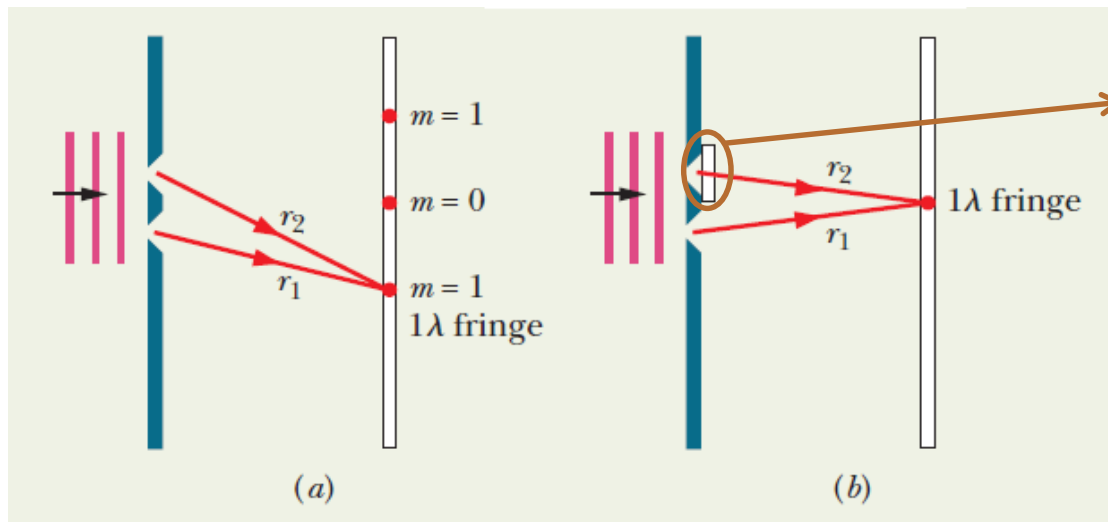
path difference (4.1)

Example: Measuring Thickness of Plastic Block

Optical path length: the physical length (L) of propagation multiplied with the corresponding refractive index (n) of the medium during propagation.

$$E = E_0 \cos(\omega t - \beta z) \text{ where } \beta z = (2\pi/\lambda_0)nz$$

Given that a plastic block of refractive index 1.5 is placed on the upper slit, determine the thickness of the plastic block.

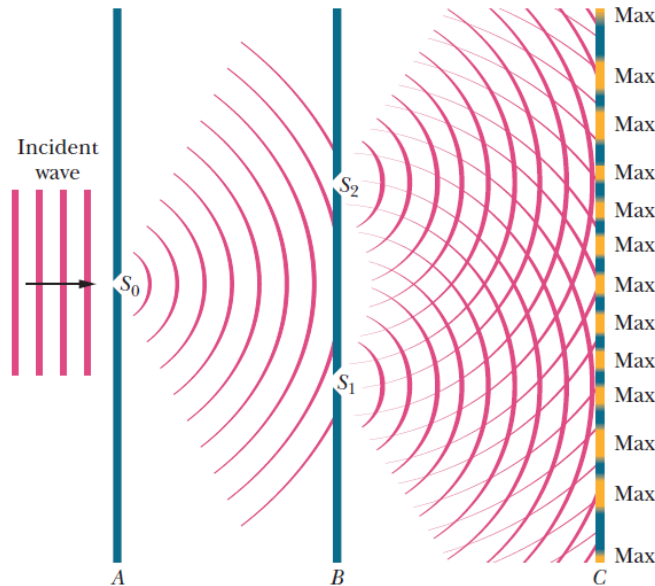


Inserted plastic block
→ the first-order bright fringe
(1λ fringe) moves to center for
 $\lambda = 600 \text{ nm}$

Let s be the plastic block thickness, the **optical path difference** at the center bright fringe is λ and is equal to $s \times (1.5 - 1)$, i.e.,

$$0.5s = 600 \times 10^{-9} \text{ m} \Rightarrow s = 1.2 \times 10^{-6} \text{ m} = 1.2 \mu\text{m}$$

Wave Coherence



- To get an **interference pattern** on the screen, the light waves reaching any point on the screen must have a **time-independent phase difference**.
- When the phase difference remains constant, the light from slits S_1 and S_2 are said to be completely **coherent**.
- If the phase difference constantly changes in time, the light is said to be **incoherent**.



Lasers

highly coherent light source



Incandescent light

incoherent light source



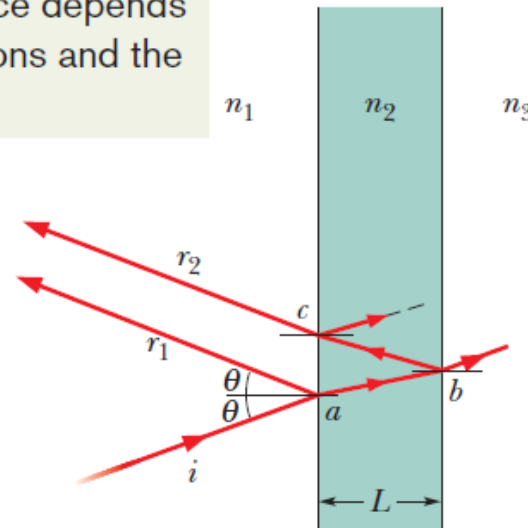
Sunlight

partially coherent: phase difference is constant only if the points under consideration are very close

Thin Film Interference



The interference depends on the reflections and the path lengths.



Ray representation of incident, transmitted, and reflected light waves.

Phase difference of the reflected waves (e.g., r_1 and r_2) is determined by:

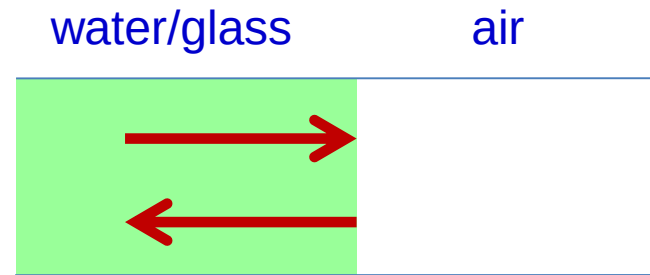
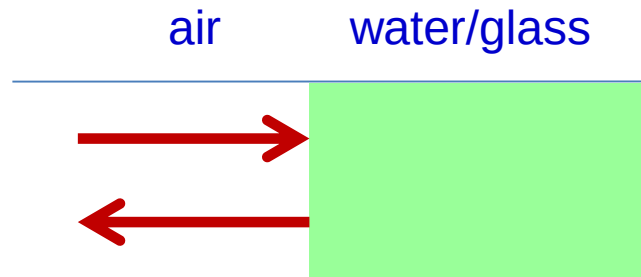
- phase shift by reflection
 - different physical path lengths
 - different refractive indices
- } different optical path lengths

Phase Shift by Reflection

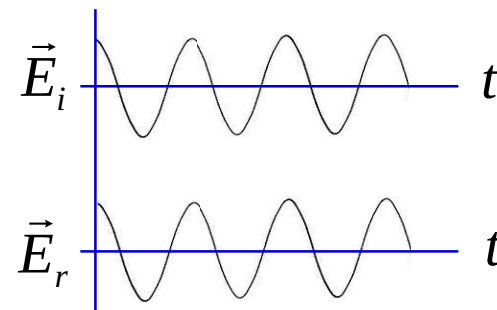
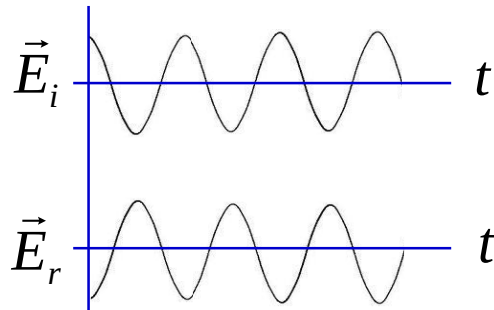
Phase shift on normal reflection (RI stands for “refractive index”):

Incident from high RI to low RI (e.g., from water/glass to air) → no phase shift

Incident from low RI to high RI (e.g., from air to water/glass) → π phase shift
i.e., $\lambda/2$ shift

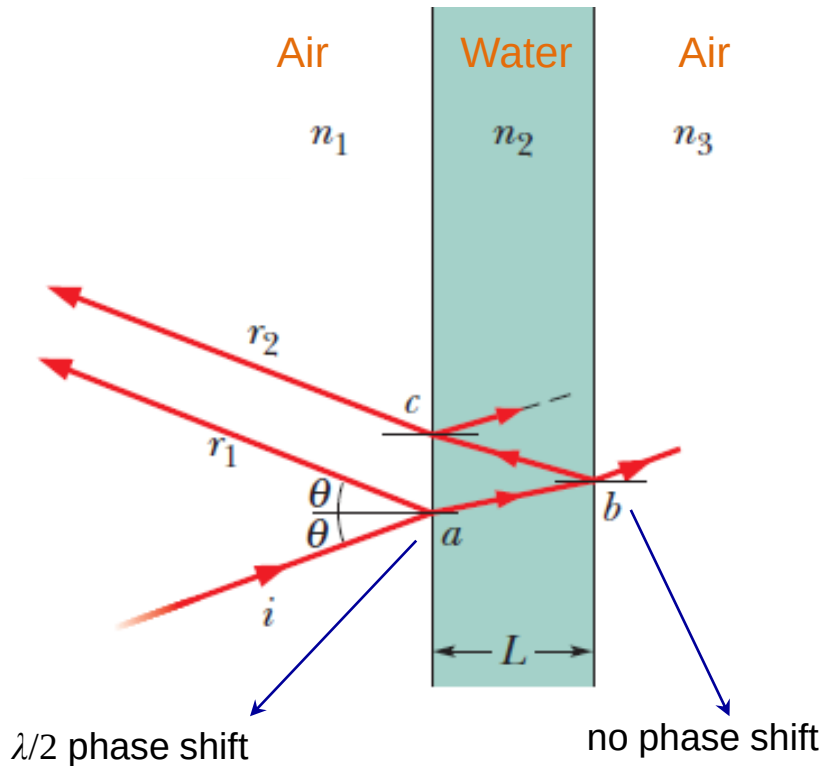


At the interface,



Please review the related contents in Topic 3.4.

Maxima and Minima



Assume normal incidence,

	r_1	r_2
Reflection phase shifts	0.5	0
	wavelength	

Path length difference	$2L$
------------------------	------

Index in which path length difference occurs	n_2
--	-------

In phase^a: $2L = \frac{\text{odd number}}{2} \times \frac{\lambda}{n_2}$

Out of phase^a: $2L = \text{integer} \times \frac{\lambda}{n_2}$

Constructive interference (Maxima):

$$2L \cdot n_2 = \left(m + \frac{1}{2} \right) \lambda, m = 0, 1, 2, \dots \quad (4.11)$$

Destructive interference (Minima):

$$2L \cdot n_2 = m\lambda, m = 0, 1, 2, \dots \quad (4.12)$$

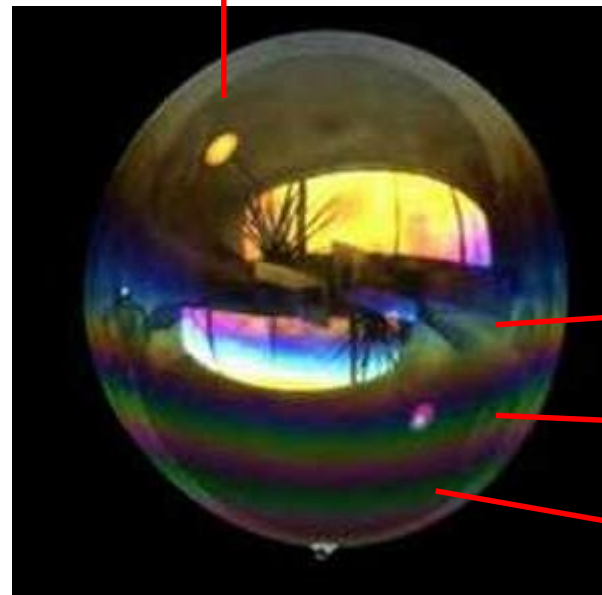
Soap Film and Bubble

Different soap film thicknesses $\rightarrow$ maxima at different wavelengths



Dark region: $L \ll \lambda$ (about to burst)

Path difference is negligible.
Phase difference is dominated by
phase shift in reflection (π)



$m = 1$

$m = 2$

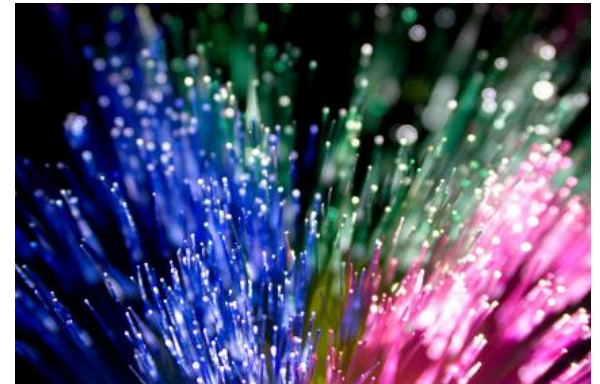
$m = 3$

Application: Thin Film Coating

Thin film coatings are widely used in glasses, camera lens, optical instruments, semiconductor optical devices, fiber devices, etc. to improve their performances through reduction or enhancement of optical reflections or transmission of specific wavelengths.

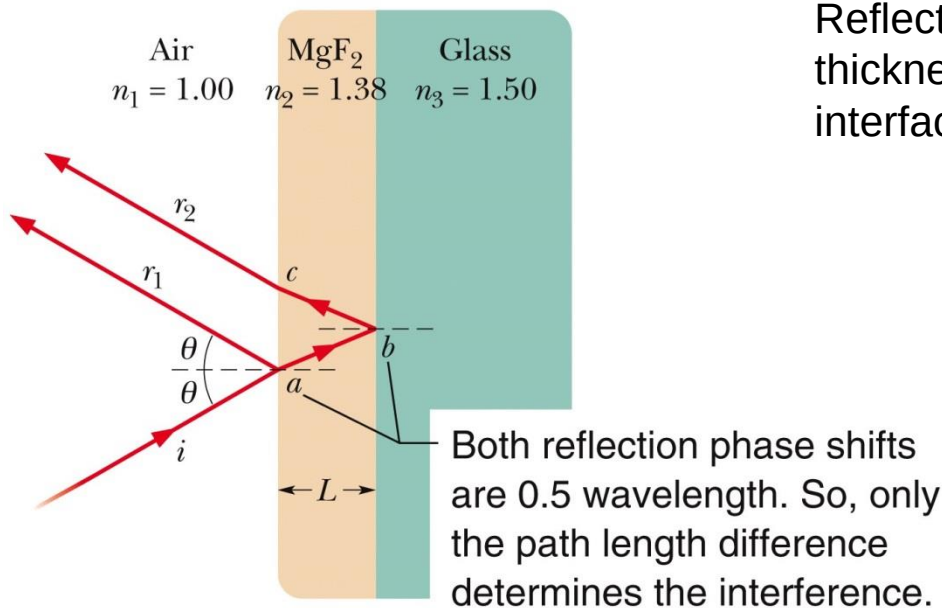


CUHK Photonic Packaging Lab
Coating Machine



Example: Antireflection Coating

A glass lens (RI = 1.50) is coated with MgF_2 (RI = 1.38) to reduce optical reflections. Determine the **minimum coating thickness** required to eliminate reflections at the center of the visible spectrum, i.e., $\lambda = 550 \text{ nm}$. Assume incident light is perpendicular to the lens surface.



Reflection is minimized if we choose coating thickness L such that waves reflected from the interfaces are exactly **out of phase**.

$$2L \cdot n_2 = \left(m + \frac{1}{2}\right)\lambda, \quad m = 0, 1, 2, \dots$$

for minimum thickness, $m = 0$

$$\Rightarrow L = \frac{\lambda}{4n_2} = \frac{550 \text{ nm}}{4 \times 1.38} = 99.6 \text{ nm}$$

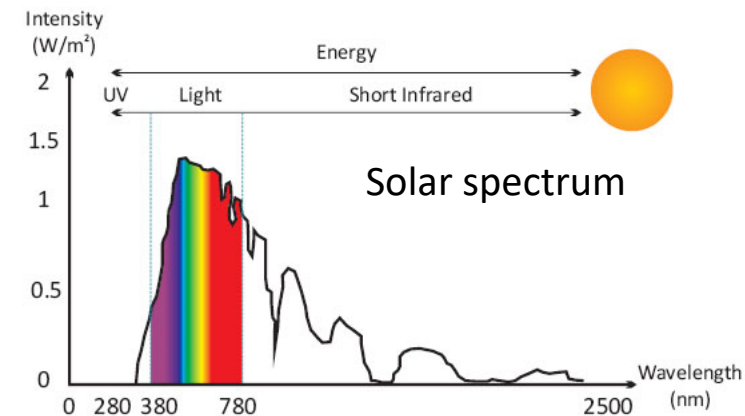
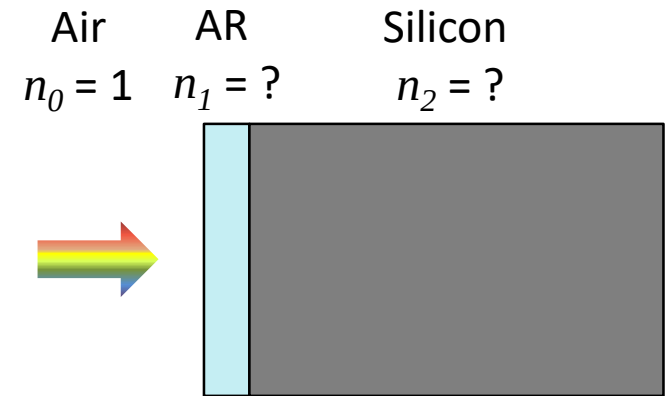
Antireflection (AR) Coating for Solar Cells

Why silicon solar cells appear to be blue?

- A thin layer of dielectric material, with a specially chosen refractive index n_1 and thickness d is coated on the solar cell.
- For normal incidence, the thickness of the AR coating $d = \lambda_0/4n_1$, where λ_0 is the wavelength of the incident light in air. (AR layer can only be optimized for certain wavelength)
- Minimum R, i.e. maximum T is achieved when

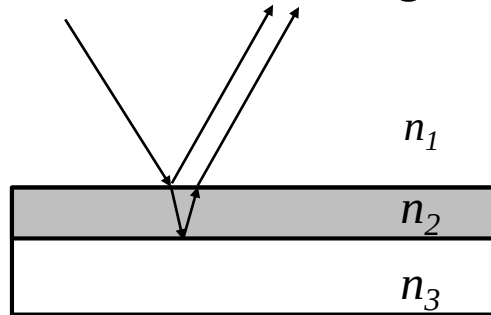
$$n_1 = \sqrt{n_0 n_2} \quad \text{Why?}$$

$$\text{Recall } r_{12} = \frac{n_1 - n_2}{n_1 + n_2} \quad T_{12} = 1 - R_{12} = 1 - r_{12}^2$$



Dielectric Mirrors

- Dielectric mirrors use a sequence of dielectric layers to obtain high reflectance at a desired wavelength

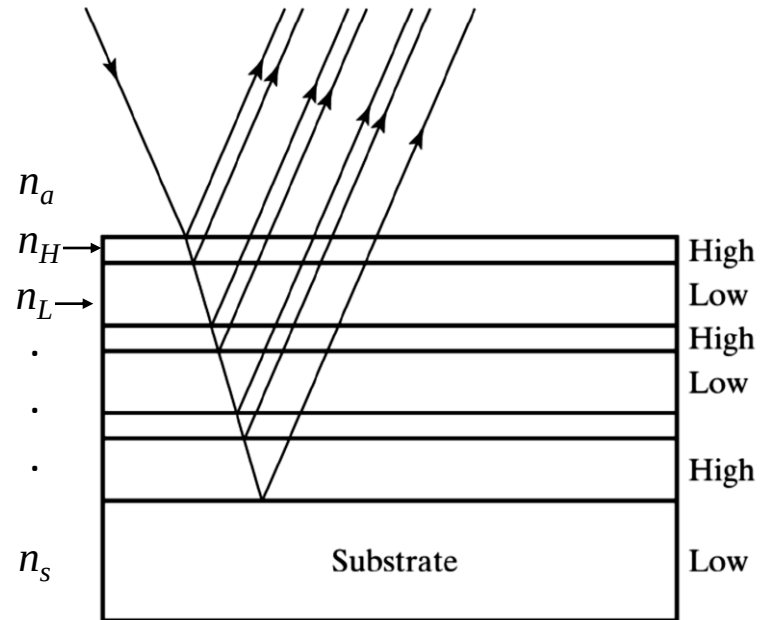


$$n_1 < n_2 \text{ and } n_2 > n_3$$

Reflection phase shift: π

- Single layer: For normal incidence, constructive interference occurs at mirror thickness $d = \lambda_0/4n_2$

- Multi-bilayer: $r_{12} = \frac{n_1 - n_2}{n_1 + n_2} \Rightarrow R_m = r_m^2 = \left(\frac{1 - \alpha}{1 + \alpha} \right)^2$ $\alpha = \frac{n_H}{n_a} \frac{n_H}{n_s} \left(\frac{n_H}{n_L} \right)^{2m}$



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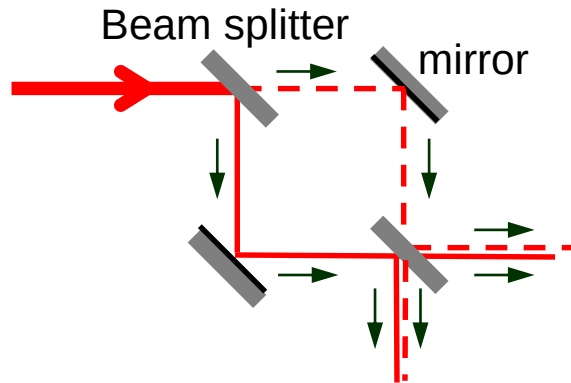
$R_m \approx 1$ when α is much larger than 1, i.e., $n_H \gg n_L$ or m is very large

- Metal mirror: absorptive, broad-band
- Dielectric mirror: non-absorptive, narrow-band (optimized thickness is λ dependent)

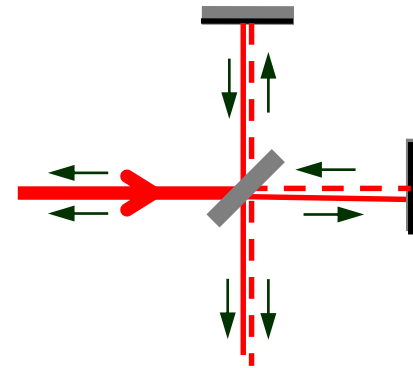
Optical Interferometer

Interferometer: an optical instrument/device that measures lengths or changes in length (or RI) with great accuracy by means of interference fringes.

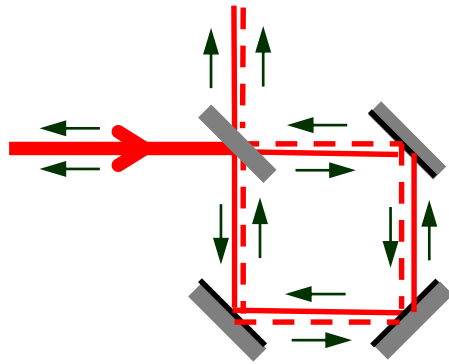
Common types of Interferometer:



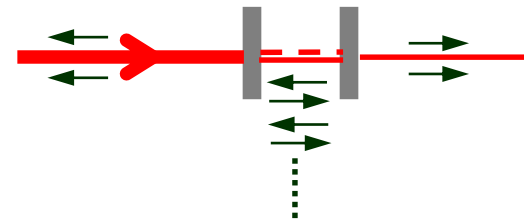
Mach-Zehnder Interferometer



Michelson Interferometer



Sagnac Interferometer



Fabry-Perot Interferometer

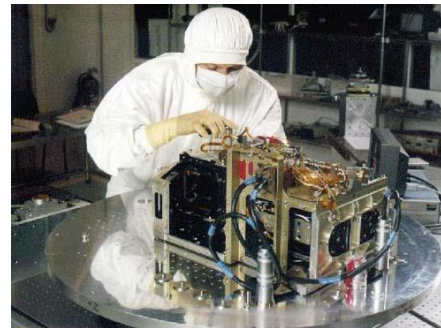
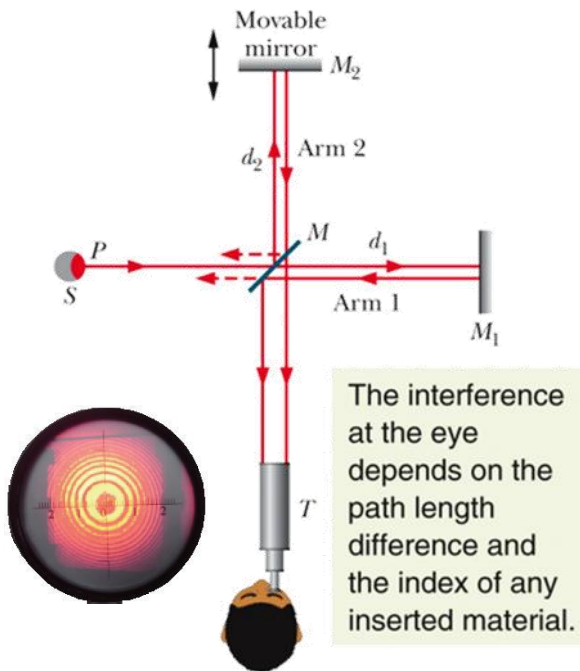
Michelson Interferometer



The Nobel Prize in Physics 1907 was awarded to Albert A. Michelson for “his optical precision instruments and the spectroscopic and metrological investigations carried out with their aid.” He was the first American to receive the Nobel Prize in Sciences.



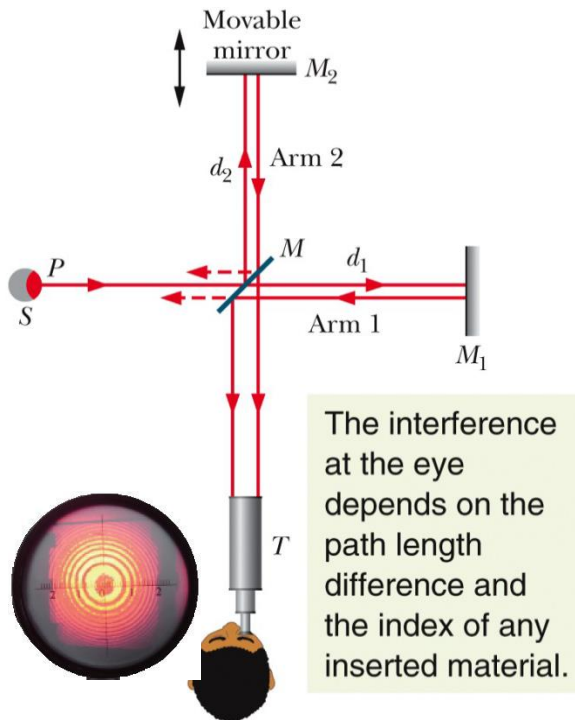
<http://en.bj-force.net/>



MIPAS Interferometer Assembly (Michelson Interferometer for Passive Atmospheric Sounding)

<https://earth.esa.int/web/guest/missions/esa-operational-eo-missions/envisat/instruments/mipas>

Michelson Interferometer



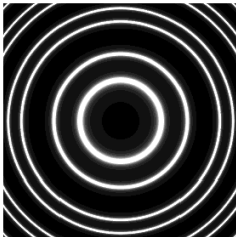
Path difference of the interfering branches is $2d_2 - 2d_1$.

As one mirror is moved by $\lambda/2$, the path difference is changed by λ and the interference pattern is shifted by one fringe.

Similarly, insertion of a transparent material of thickness L and RI n on one of the paths causes a phase change of

$$\Delta\phi = \frac{2\pi}{\lambda} [2L(n - 1)] \quad (4.13)$$

For each path change of λ (phase change = 2π), the pattern is shifted by one fringe. Thus, by counting the number of fringes through which the material causes the pattern to shift, the thickness L can be determined in terms of λ .



Animation on shifting of interference fringes

<http://skullsinthestars.com/2008/10/16/fabry-perot-and-their-wonderful-interferometer-1897-1899/>

▼ About the Nobel Prize in
Physics 2017

Summary

Prize Announcement

Press Release

Advanced Information

Popular Information

Award Ceremony Video

Award Ceremony Speech

► Rainer Weiss

► Barry C. Barish

► Kip S. Thorne

All Nobel Prizes in Physics

All Nobel Prizes in 2017

The Nobel Prize in Physics 2017



Photo: Bryce Vickmark

Rainer Weiss

Prize share: 1/2

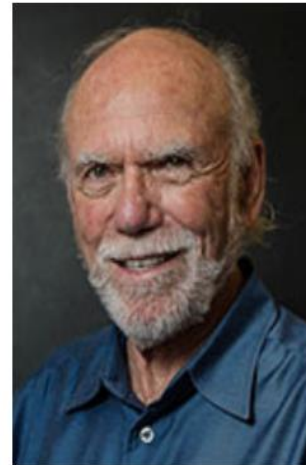


Photo: Caltech

Barry C. Barish

Prize share: 1/4



Photo: Caltech Alumni
Association

Kip S. Thorne

Prize share: 1/4

The Nobel Prize in Physics 2017 was divided, one half awarded to Rainer Weiss, the other half jointly to Barry C. Barish and Kip S. Thorne *"for decisive contributions to the LIGO detector and the observation of gravitational waves"*.

https://www.nobelprize.org/nobel_prizes/physics/laureates/2017/

Application: Gravitational Waves Detection



**Gravitational Waves Detected 100 Years After
Einstein's Prediction**

News Release • February 11, 2016

<https://www.ligo.caltech.edu/news/ligo20160211>

*LIGO Opens New Window on the Universe with
Observation of Gravitational Waves from Colliding Black Holes*

WASHINGTON, DC/Cascina, Italy

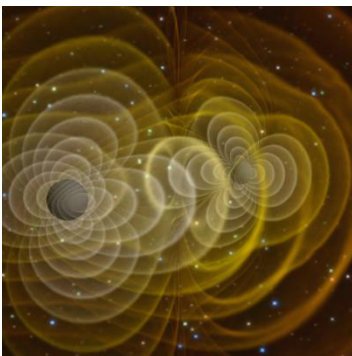
For the first time, scientists have observed ripples in the fabric of spacetime called gravitational waves, arriving at the earth from a cataclysmic event in the distant universe. This confirms a major prediction of **Albert Einstein's 1915 general theory of relativity** and opens an unprecedented new window onto the cosmos. Gravitational waves carry information about their dramatic origins and about the nature of gravity that cannot otherwise be obtained. Physicists have concluded that the **detected gravitational waves were produced during the final fraction of a second of the merger of two black holes to produce a single, more massive spinning black hole.** This collision of two black holes had been predicted but never observed.

The gravitational waves were **detected on September 14, 2015** at 5:51 a.m. Eastern Daylight Time (09:51 UTC) by both of the **twin Laser Interferometer Gravitational-wave Observatory (LIGO) detectors**, located in Livingston, Louisiana, and Hanford, Washington, USA. The

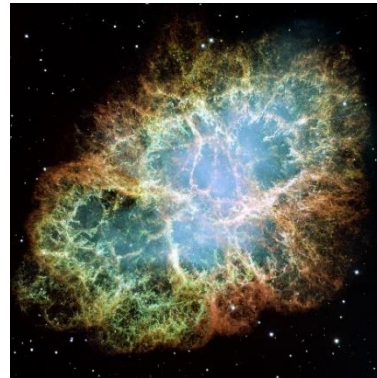
Gravitational Waves

What are gravitational waves? Gravitational waves are 'ripples' in the fabric of space-time caused by some of the most powerful processes in the universe - **colliding black holes, exploding stars, and even the birth of the universe itself**. Albert Einstein predicted the existence of gravitational waves in 1916, derived from his general theory of relativity. Einstein's mathematics showed that **massive accelerating objects** (such as neutron stars or black holes orbiting each other) would **disrupt space-time in such a way that waves of distorted space would radiate from the source**. These ripples travel at the speed of light through the universe, carrying information about their origins, as well as clues to the nature of gravity itself.

Detecting and analyzing the information carried by gravitational waves will allow us to observe universe in a way never before possible. This will open up a new window on the universe. <http://www.ligo.org/science/faq.php>



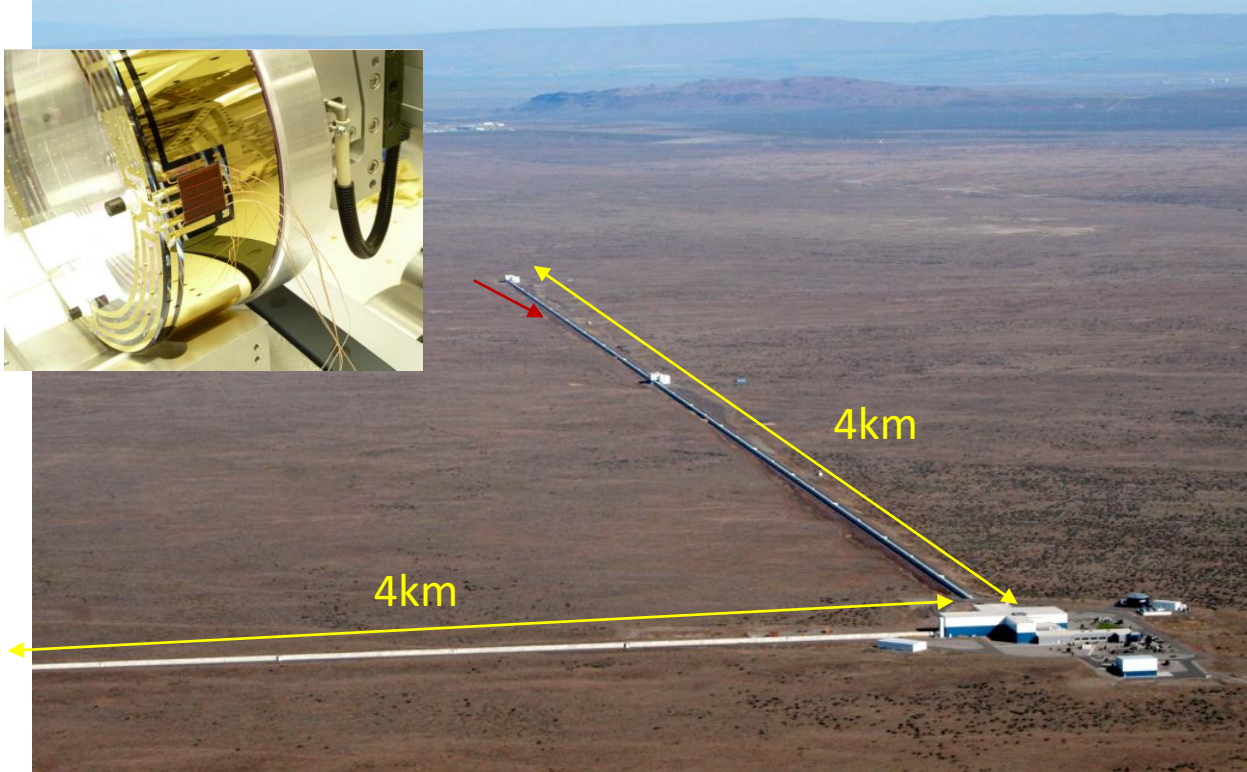
Gravitational waves produced by 2 orbiting black holes. Image: Henze, NASA



Crab Nebula supernova remnant. Such a core-collapse supernova could produce a burst of gravitational waves. Image: Hester, Loll, NASA

Laser Interferometer Gravitational-Wave Observatory (LIGO)

<https://www.ligo.caltech.edu/WA/page/facts>

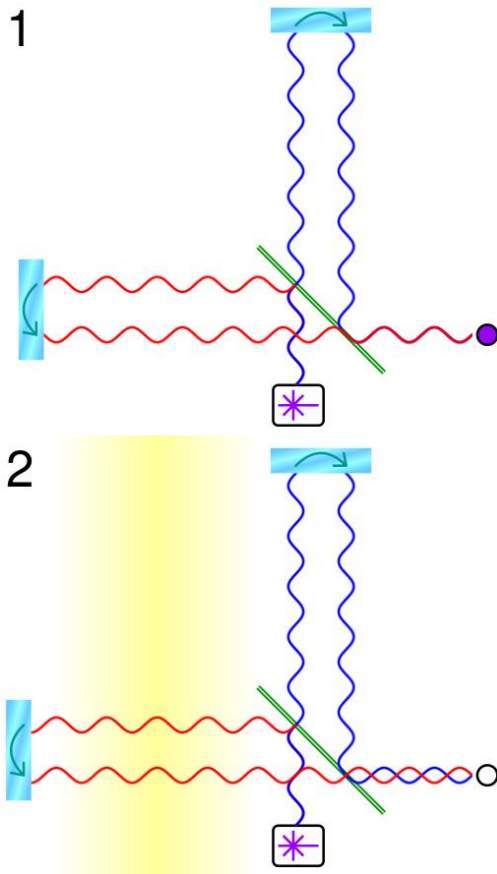


LIGO at Hanford, USA



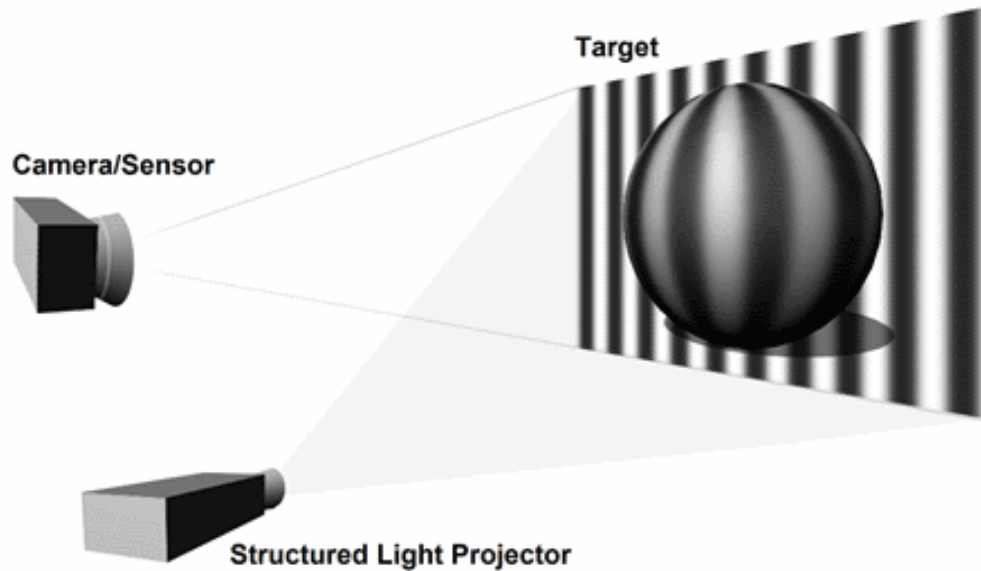
- Two Michelson interferometers with 4 km long arms in vacuum, 3000 kilometers apart
 - Capability to measure changes in length 10,000 times *smaller* than an atomic nucleus
- When a gravitational wave passes by, the stretching and squashing of space causes the arms of the interferometer alternately to lengthen and shorten, one getting longer while the other gets shorter and then vice-versa.

Detection of Gravitational Waves



Simplified operation of a gravitational wave observatory **Figure 1:** A beamsplitter (green line) splits coherent light (from the white box) into two beams which reflect off the mirrors (cyan oblongs); only one outgoing and reflected beam in each arm is shown, and separated for clarity. The reflected beams recombine and an interference pattern is detected (purple circle). **Figure 2:** A gravitational wave passing over the left arm (yellow) changes its length and thus the interference pattern. (Source: Wikipedia)

Application: Structured Light Imaging

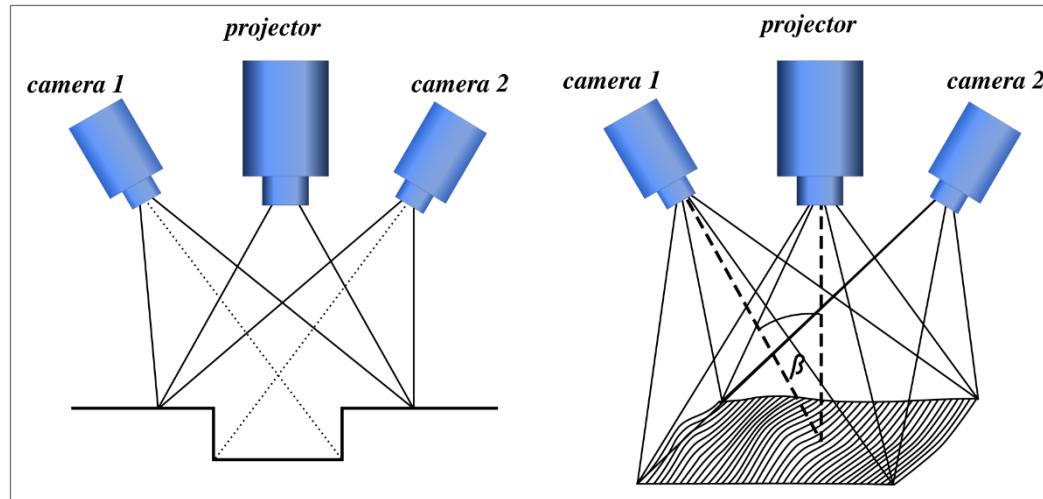


Structured Light Imaging captures the 3-D topography of a surface using specific patterns of light, and a 2D imaging camera.

Principle: a known (interference) pattern is projected onto a surface. When the camera views the pattern, the surface features of the target distort the pattern. The direction and size of the pattern distortions are used to reconstruct the surface topography of the target object.

<https://www.robotictomorrow.com/article/2018/04/what-is-structured-light-imaging/11821>

Application: Structured Light Imaging



Fringe pattern recording system with 2 cameras (avoiding obstructions)

The *laser interference method* works with two wide planar laser beam fronts. Their interference results in regular, equidistant line patterns. Different pattern sizes can be obtained by changing the angle between these beams. The method allows for the exact and easy generation of very fine patterns with unlimited depth of field.

https://en.wikipedia.org/wiki/Structured-light_3D_scanner

Application: Structured Light Imaging



The right half image shows the subject being captured and the left image shows the reconstructed 3D geometry.

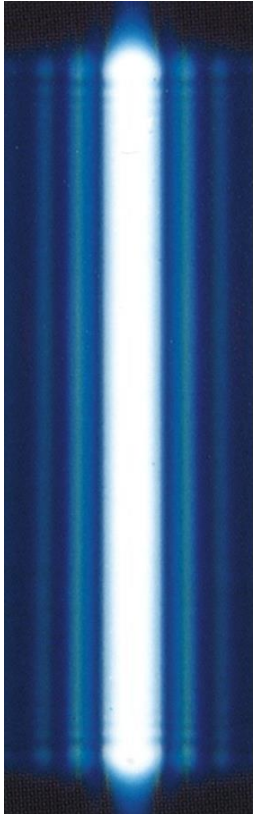


The structured light system, which Apple debuted on the iPhone X, consists of a proximity sensor that tells the system if the user is within the correct distance and a sensor that evaluates whether to further illuminate the user if the environment is too dark, a dot projector that projects a pattern of 30,000 dots on the users face an infrared camera that takes a photo of the dot pattern.

S. Zhang, "High-speed 3D shape measurement with structured light methods: A review," Optics and Lasers in Engineering, vol.106, p.119-131, 2018.

https://www.oled-a.org/apple-to-replace-structured-light-system-wtof_12020.html

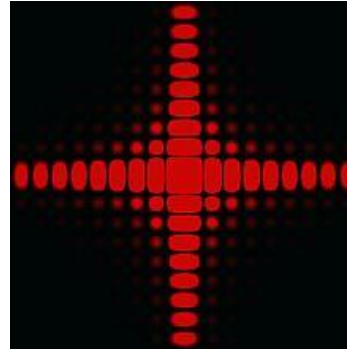
Diffraction Phenomena



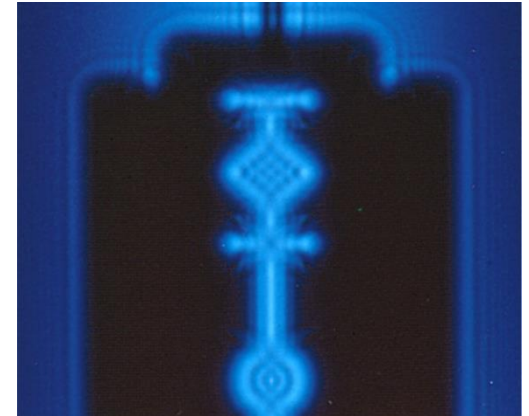
vertical slit



circular disk



square aperture



razor blade



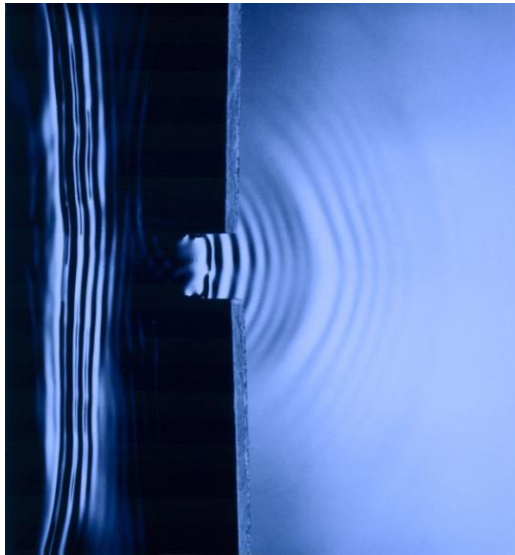
diffraction photography



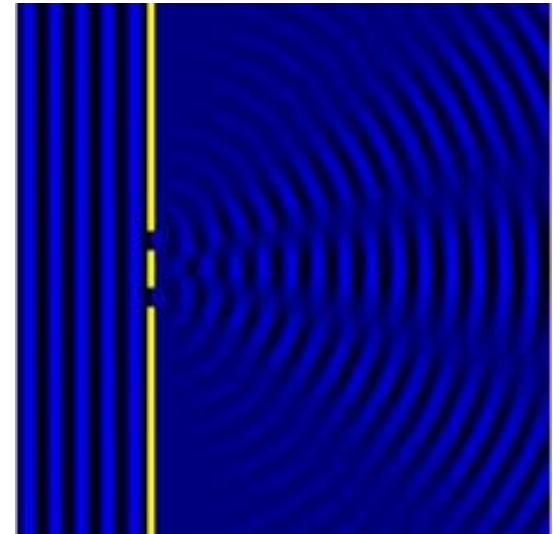
sunlight diffraction

Basics of Diffraction

Light diffraction: light passes through a narrow slit (dimension $\sim$ wavelength) or an edge **interferes with itself and produces an interference pattern called the diffraction pattern**. It demonstrates the **wave nature** of light.



- Water wave encounters a barrier with an opening of dimensions similar to the wavelength, the part of the wave that passes through the opening will flare and spread out (i.e., diffract) into the region beyond the barrier.
- The flaring is consistent with the spreading of wavelets according to Huygens principle.



Interference pattern from two-slit diffraction

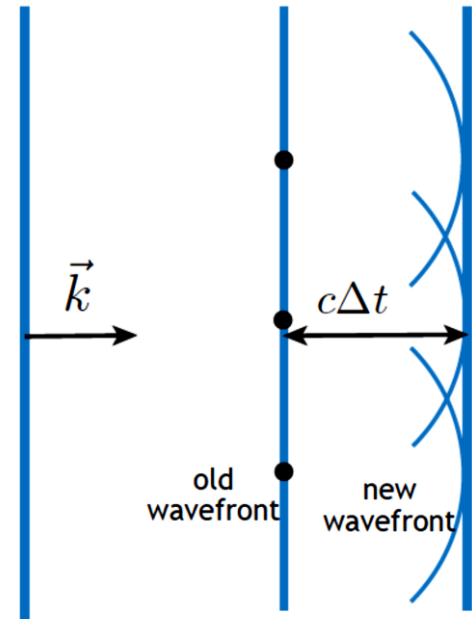
<http://en.wikipedia.org/wiki/Diffraction>

Huygens' Principle

- Huygens' Principle is a geometric construction for **determining the position of a new wave** at some point based on the knowledge of the wave front that preceded it
- **All points on a given wave front are taken as point sources for the production of spherical secondary waves**, called wavelets, which propagate in the forward direction with speeds characteristic of waves in that medium

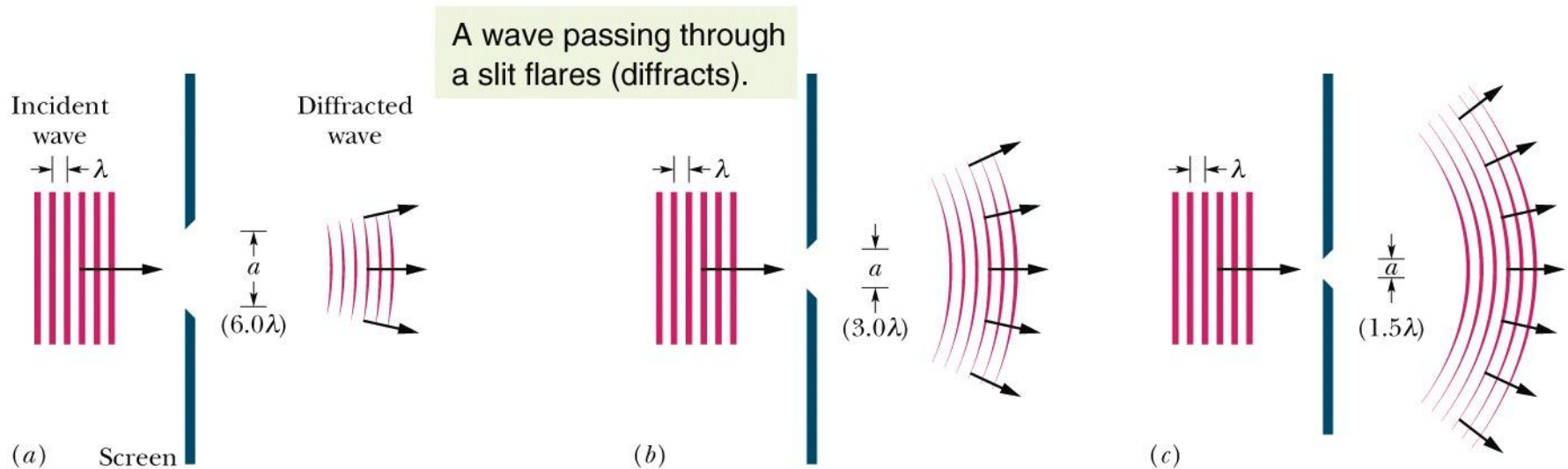
Construction for a Plane Wave

- The points are representative sources for the wavelets
- After the wavelets have moved a distance $c\Delta t$, a new plane can be drawn tangent to the wavefronts



(Source: MIT Opencourseware 6.007)

Dependence on Slit Width

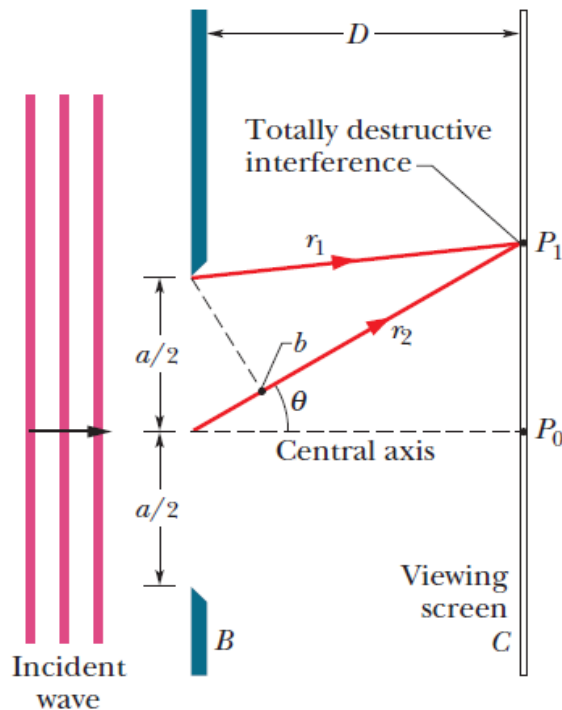


The narrower the slit, the stronger the diffraction!

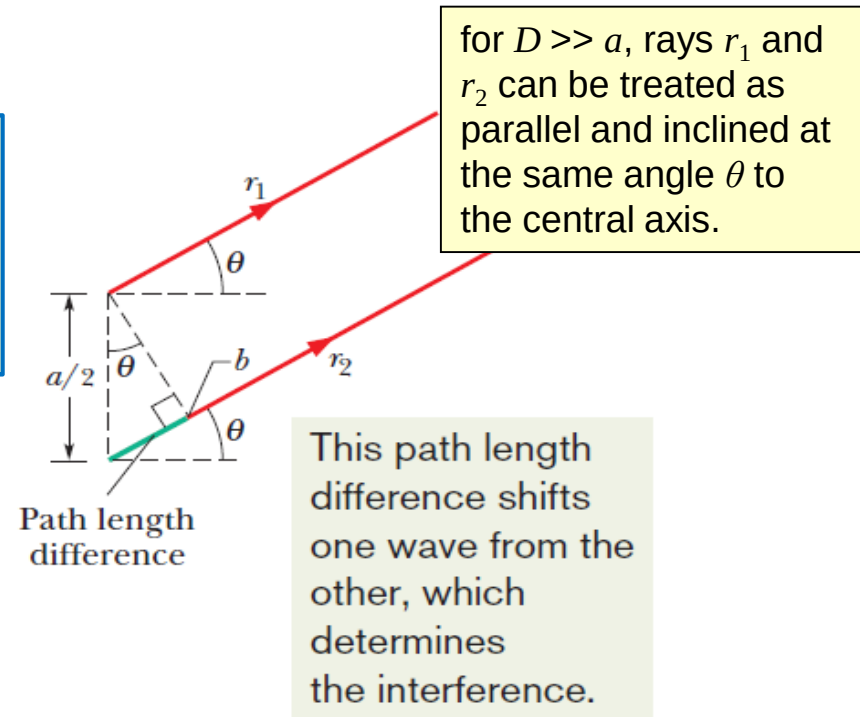
Diffraction limits geometrical optics (ray optics) since a narrow slit will cause the light to spread. Geometrical optics holds only when slits or apertures have dimensions much greater than the wavelength of light.

When projecting an image of tiny features on a photo film, the image will become blurred through optical diffraction.

Single Slit Diffraction: Locating Minima



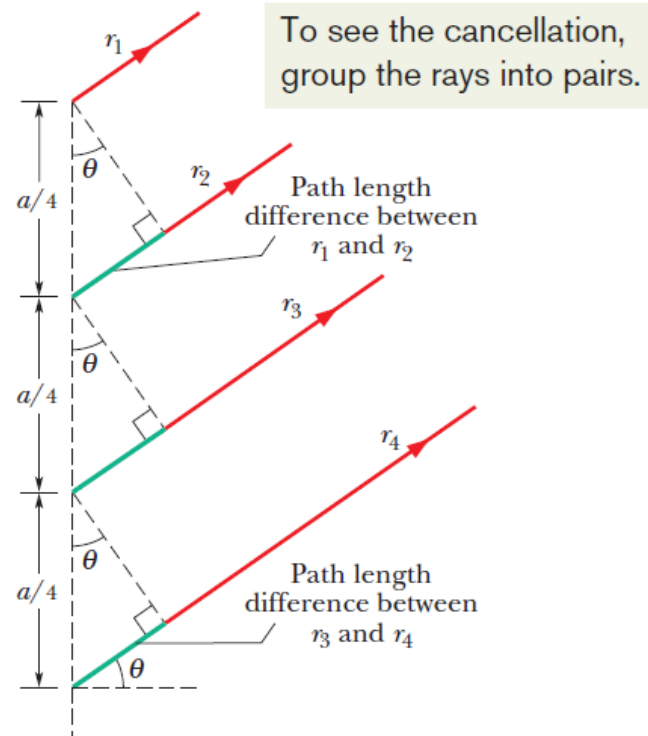
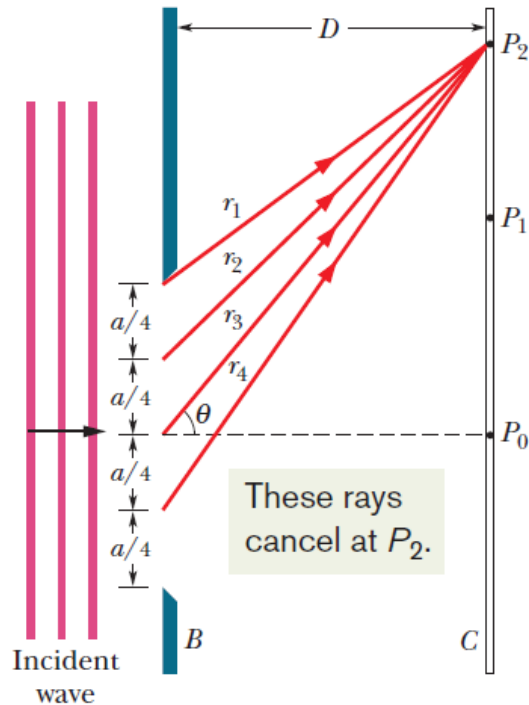
This pair of rays r_1 and r_2 must cancel each other at the minimum P_1 . Similarly, we can pair up other rays in the two zones.



- We divide the slit into two zones of equal widths $a/2$ and then consider a light ray r_1 from the top point of the top zone and a light ray r_2 from the top point of the bottom zone.
- For destructive interference at P_1 , $\frac{a}{2} \sin\theta = \frac{\lambda}{2}$. Hence, the 1st minimum is located at

$$a \sin\theta = \lambda \quad (4.14)$$

Single Slit Diffraction: Locating Minima



- Following the same principle in the last slide, one can find the 2nd dark fringe P_2 . Now we should divide the slit into four zones of equal widths $a/4$. The 2nd minimum is located at $(a/4) \sin \theta_2 = \lambda/2$, i.e., $a \sin \theta_2 = 2\lambda$
- Likewise, the m th minimum (dark fringe) is located at $(a/2m) \sin \theta_m = \lambda/2$

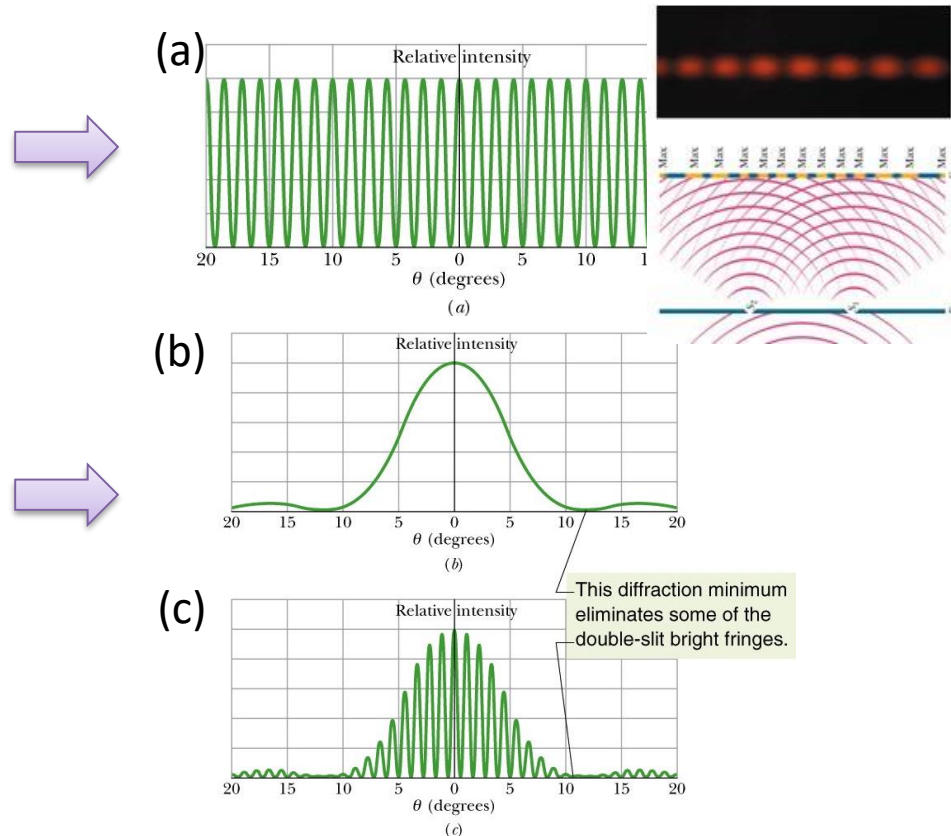
$$a \sin \theta_m = m\lambda, \quad \text{for } m = 1, 2, 3, \dots \quad (4.15)$$

Double-Slit Diffraction

Combined effect of double-slit interference and single-slit diffraction.

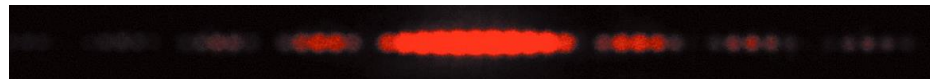
When slit width $a \ll \lambda \rightarrow$ reduce to double-slit interference. Central maximum of the diffraction pattern of each slit covers the entire screen. Interference leads to fringes of same intensities.

If $a \ll \lambda$ is not satisfied $\rightarrow$ each slit produces its own diffraction pattern described previously. The interference of light from the two slits produces intensity fringes. The diffraction pattern in (b) sets an envelope for the intensity plot in (a) to produce plot (c).



Recorded images

Double-slit diffraction

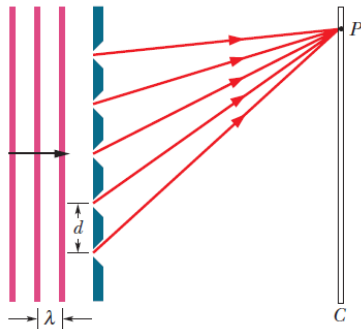


Single-slit diffraction



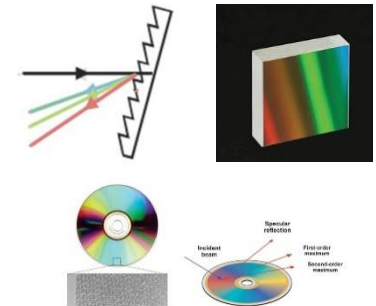
Diffraction Gratings

Diffraction gratings contain a large number of slits, often called **rulings**. They can be made on an opaque surface with narrow parallel grooves arranged like the slits, forming a **reflection grating** instead of a **transmission grating**.

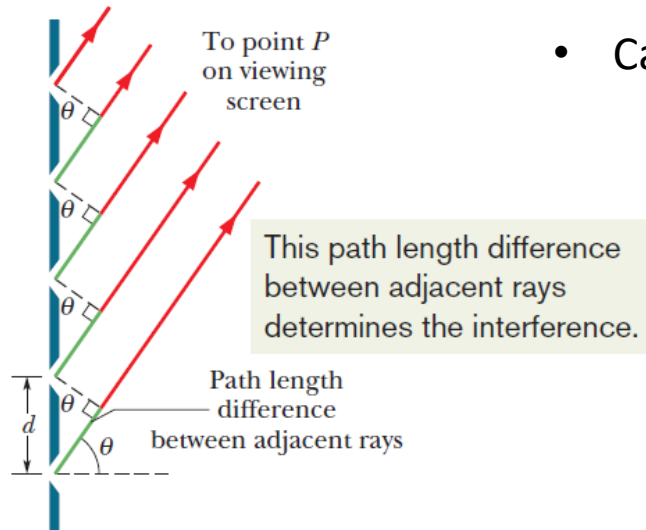


- Slits still separated by distance d
- Maxima again occur only for

$$d \sin \theta_m = m \lambda \quad \text{for } m = 0, 1, 2, \dots \quad (4.16)$$



reflection gratings



- Calculation of pattern intensity

$$I_{total} = I_0 \left[\frac{\sin \alpha}{\alpha} \right]^2 \left[\frac{\sin \left(\frac{M \Delta \phi}{2} \right)}{\sin \left(\frac{\Delta \phi}{2} \right)} \right]^2$$

Diffraction
Interference

$$\alpha = \pi a \sin \theta / \lambda$$

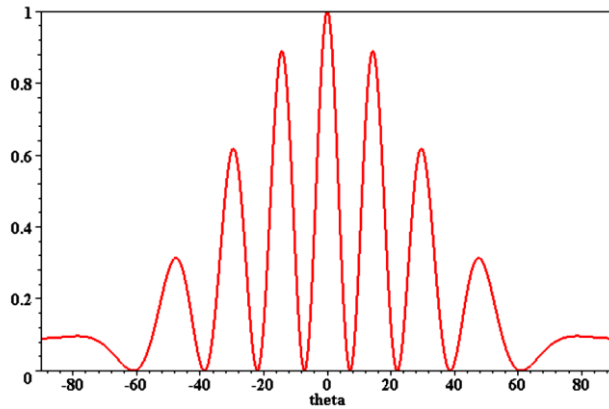
$$\Delta \phi = 2 \pi d \sin \theta / \lambda$$

Diffraction Gratings

Interference fringe width is inversely proportional to the number N of rulings for a given ruling separation d and wavelength λ .

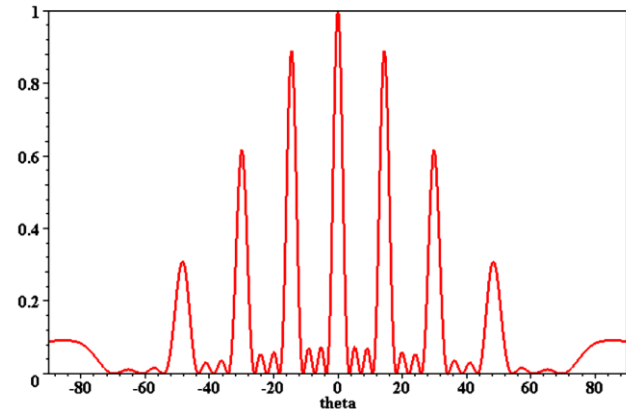
2 Slits: $d = 4\lambda$

Double slit, $d=4\lambda$



4 Slits: $d = 4\lambda$

Grating: $N=4$, $d=4\lambda$



10 Slits: $d = 4\lambda$

Grating: $N=10$, $d=4\lambda$

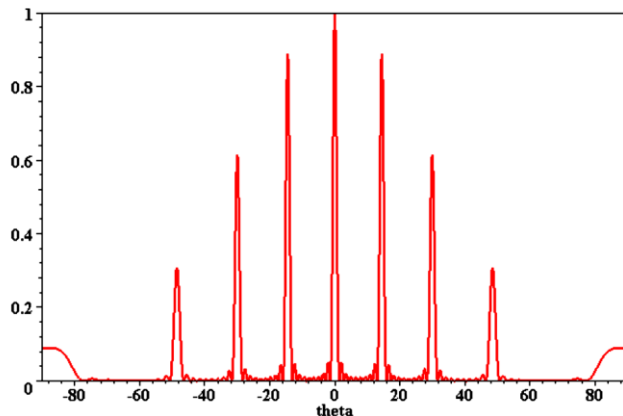
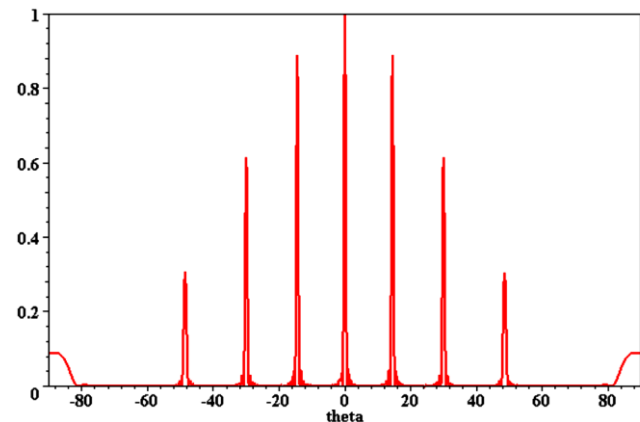


Figure source:

<http://www.phys.ufl.edu/courses/phy2049/>

20 Slits: $d = 4\lambda$

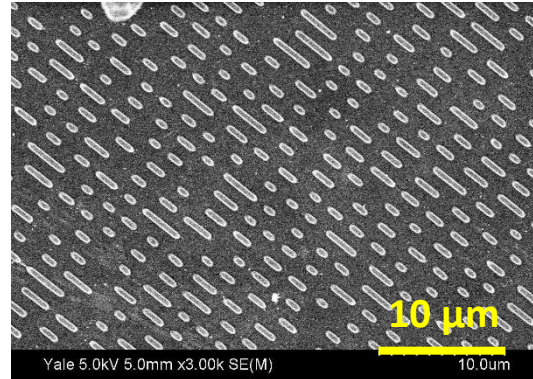
Grating: $N=20$, $d=4\lambda$



Applications of Diffraction Gratings

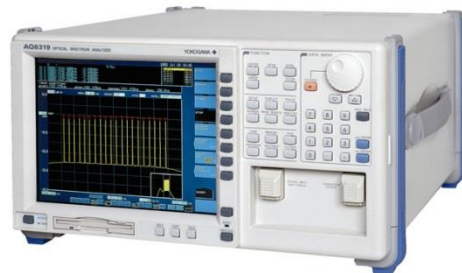


Compact disk (CD)

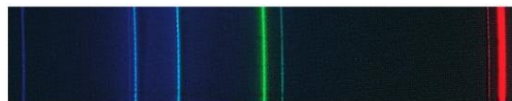


SEM image of CD surface

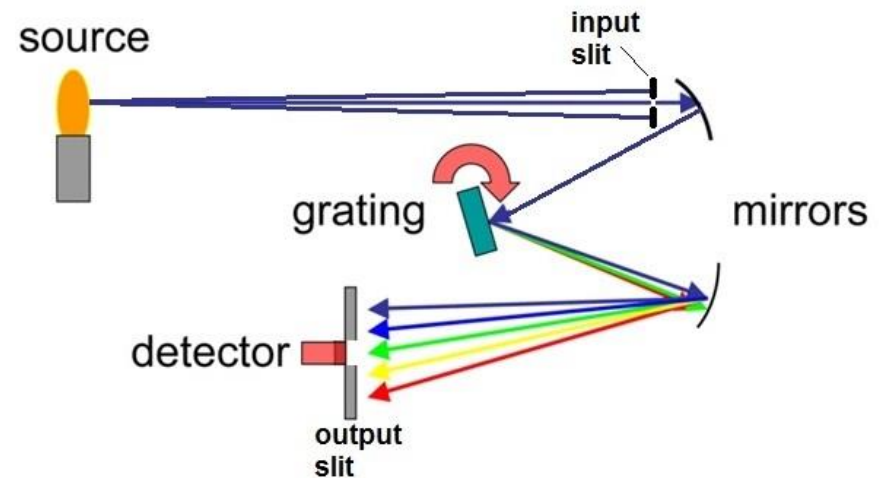
Each tick is 1 μm . The fine rulings of $\sim 0.5 \mu\text{m}$ width on a CD function serve as a reflection grating



Optical spectrum analyzer (OSA)
manufactured by Yokogawa

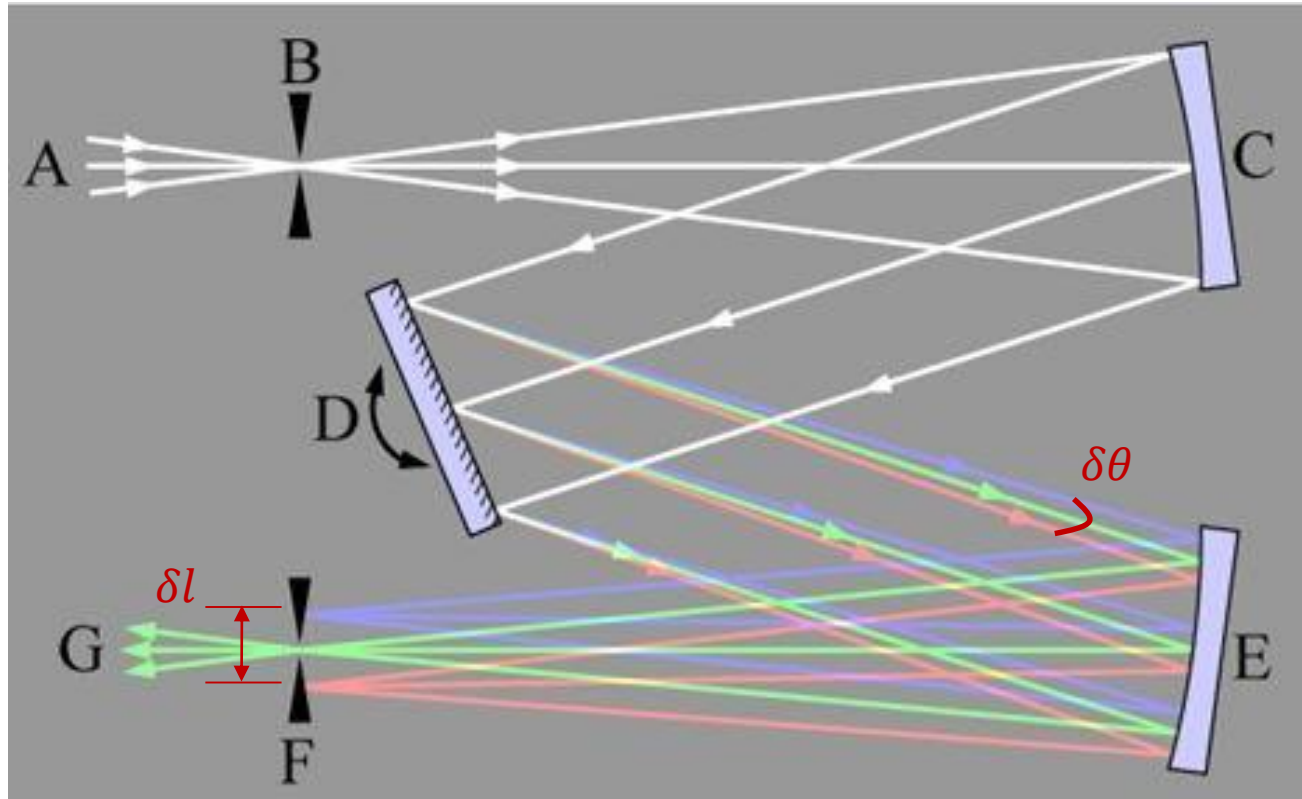


Visible emission lines of Cadmium resolved with an OSA

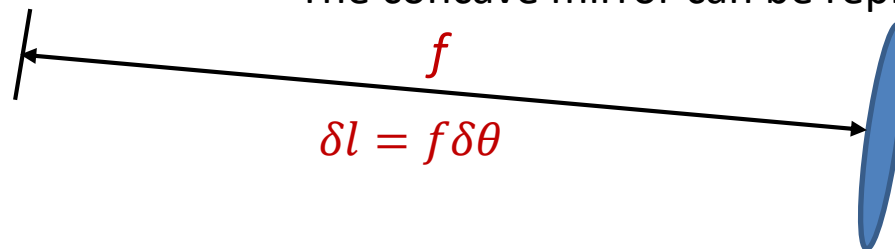


the inside of an OSA

Example: Optical Spectrum Analyzer



The concave mirror can be replaced by a convex lens



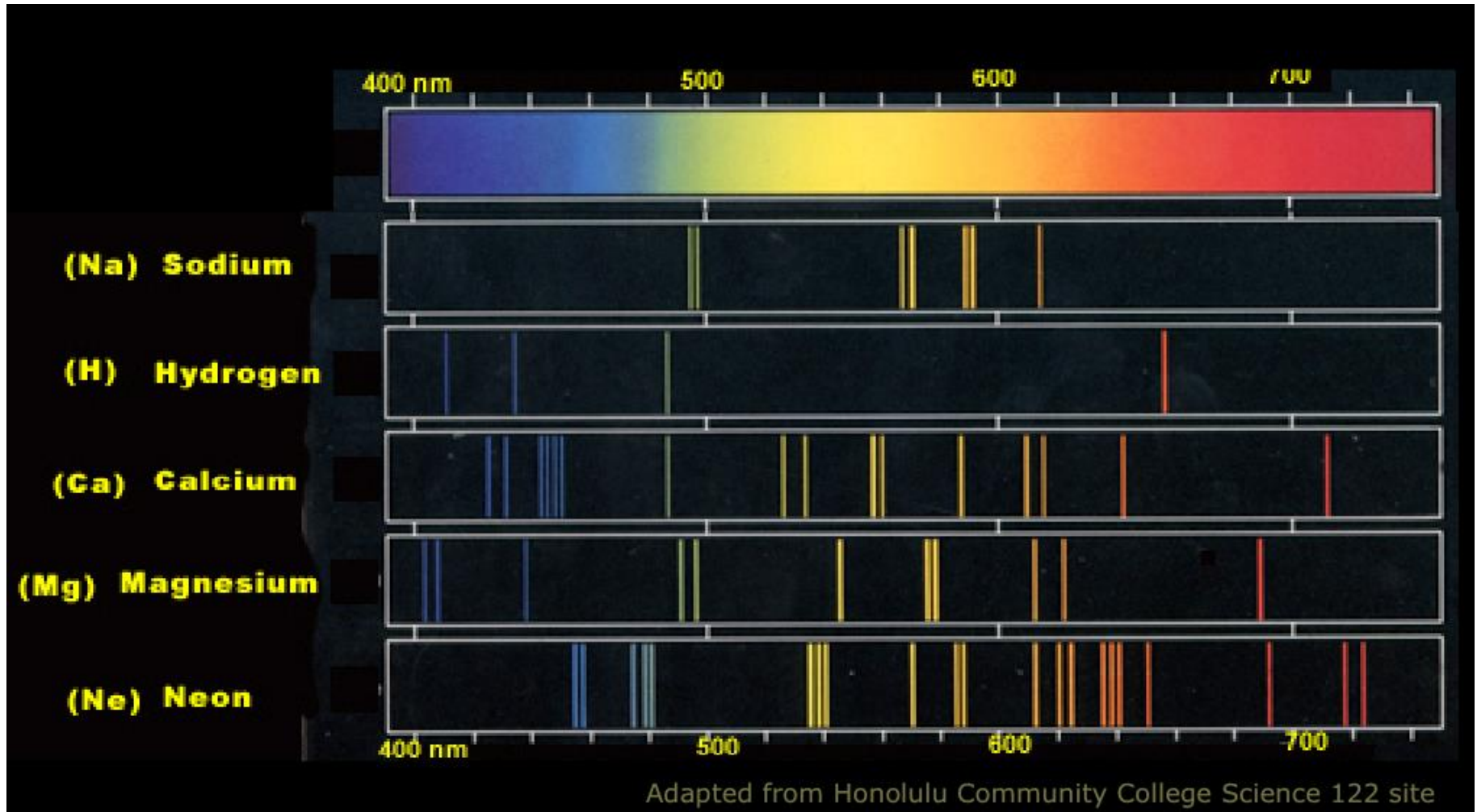
Example: Optical Spectrum Analyzer

Grating-based optical spectrum analyzer (OSA) relies on diffraction grating to **resolve light of different wavelengths**. One of the most important specifications of a spectrum analyzer is its dispersive power, which characterizes the **angular separation $\delta\theta$ or linear separation δl** for spectrum lines for wavelength difference of $\delta\lambda$. **angular dispersive power is $D_\theta \equiv \frac{\delta\theta}{\delta\lambda}$** , and for **linear dispersive power is $D_l \equiv \frac{\delta l}{\delta\lambda}$** . The sodium yellow light contains two wavelengths at $\lambda = 589.00 \text{ nm}$ and $\lambda' = 589.59 \text{ nm}$. Find out their position and angular separation when analyzed using the **1st-order diffraction** of a grating-based OSA with a grating that is **15.0 cm** long and has the number of rulings **180,000**. Assume the **focal length** of the objective lens of the OSA is **1.0 m**.

- $d = \frac{15.0\text{cm}}{180000} = \frac{1}{12000} \text{ cm}$, so the angular position $\theta = \sin^{-1} \frac{\lambda}{d} = \sin^{-1} 0.7068 = 44.975^\circ$.
- Differentiate both sides of (4.16): $d \sin \theta_m = m\lambda$, we have $d \cos \theta_m \delta\theta = m\delta\lambda$, therefore $D_\theta \equiv \frac{\delta\theta}{\delta\lambda} = \frac{m}{d \cos \theta_m}$ We have $D_\theta = \frac{1}{(\frac{1}{12000}\text{cm}) \cos 44.975^\circ} = 1.696 \times 10^{-3} \text{ rad/nm}$
Therefore, the angular separation of the double line in the spectrum is $\delta\theta = D_\theta \delta\lambda = 1.696 \times 10^{-3} \text{ rad/nm} \times 0.59 = 10^{-3} \text{ rad}$.
- The linear separation of the double line is $\delta l = f \delta\theta = 1 \text{ mm}$.

Distinguishing Element “Signatures”

- Application of Optical Spectrum Analyzer



Diffraction by Circular Aperture

- A tiny circular aperture such as a **circular lens** can result in an optical diffraction pattern. (diffraction pattern is viewed at a long distance from the diffracting object, and also when it is viewed at the focal plane of an imaging lens)
- Shown on the right is the diffraction pattern of a circular aperture. You can find the central maximum and the interference fringes. For an aperture of **diameter d (similar to slit width)**, the first minimum is located at an angle given by

$$\sin\theta = 1.22 \frac{\lambda}{d} \quad (4.17)$$

Compared to (4.14), the factor 1.22 here is introduced by the circular shape of the aperture.

- In photography, a larger aperture (smaller f -number) gives a better resolved image due to weaker diffraction effect (i.e. smaller θ):



f/5.6



f/22



f/32



f/45

<http://www.luminous-landscape.com/tutorials/understanding-series/u-diffraction.shtml>

Numerical Aperture and Airy Disc Size

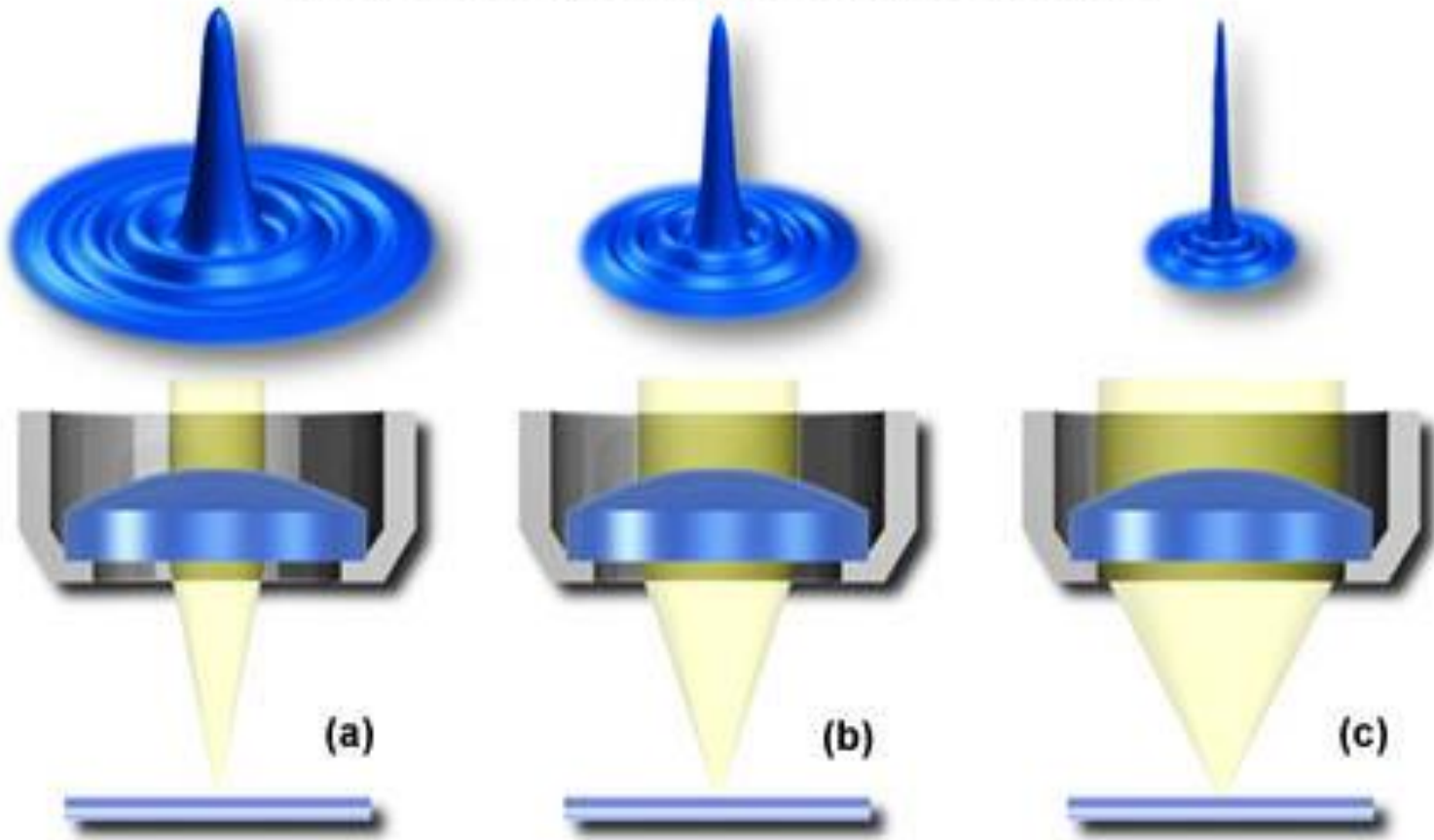
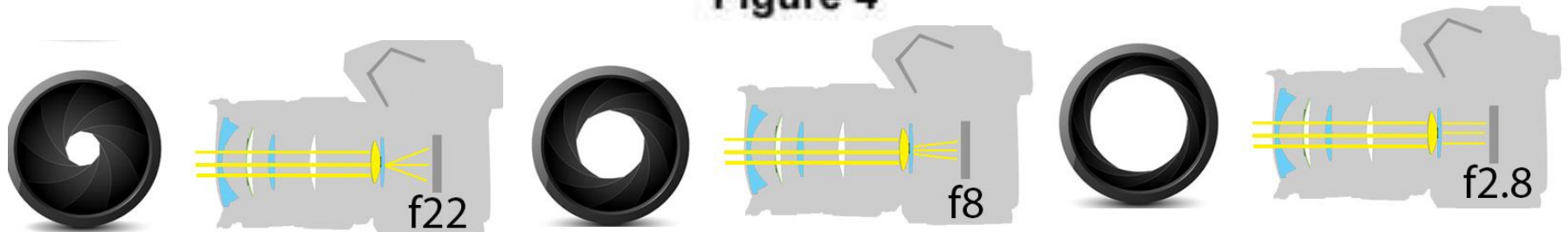


Figure 4

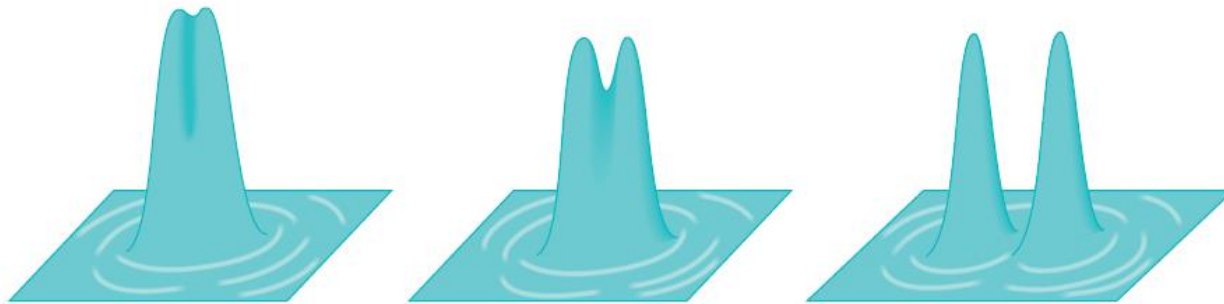


<https://micro.magnet.fsu.edu/primer/anatomy/numaperture.html>

<https://fstoppers.com/studio/fstoppers-original-what-lens-diffraction-and-when-does-diffraction-happen-6022>

Resolvability

Consider the lens image of two distant point objects (e.g., stars) with a small angular separation. They cannot be resolved (distinguished) if their diffraction patterns overlap. Two objects are barely resolvable when their angular separation is given by $\theta_R \approx \sin\theta_R = 1.22 \frac{\lambda}{d}$. That is, the **central maximum of one source coincides with the first minimum of the other source** in the diffraction pattern. This is called the **Rayleigh's criterion**.



Indistinguishable marginally distinguishable clearly distinguishable

Diffraction Limit in Optical Imaging

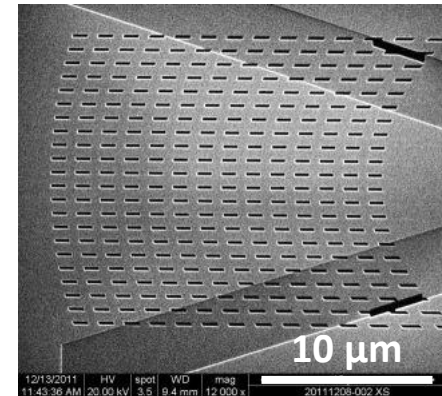
- **Diffraction Limit:** a fundamental limit to the resolution of an imaging system, such as camera, telescope, and microscope.
- The resolution is proportional to the size of the objective lens, and inversely proportional to the wavelength. The **resolution limit is roughly equal to half of the wavelength**. For example, using green light with wavelength $0.5\ \mu\text{m}$ for imaging, the resolution limit is $\sim 0.25\ \mu\text{m}$.
- To increase the resolution, shorter wavelengths such as **ultraviolet or X-ray** are used for imaging. Alternatively, instead of using optical imaging, one can use **electronic imaging**. Electrons exhibit wave-particle duality with a **wavelength (de Broglie wavelength) inversely proportional to its momentum**, which can reach $\sim 0.01\ \text{nm}$ for $10\ \text{keV}$ energy.



Optical microscope (Leica)



SEM (JEOL)



SEM image of subwavelength nanofeatures fabricated on silicon at CUHK (Courtesy of Prof. H. K. Tsang)

TELESCOPE IMAGING

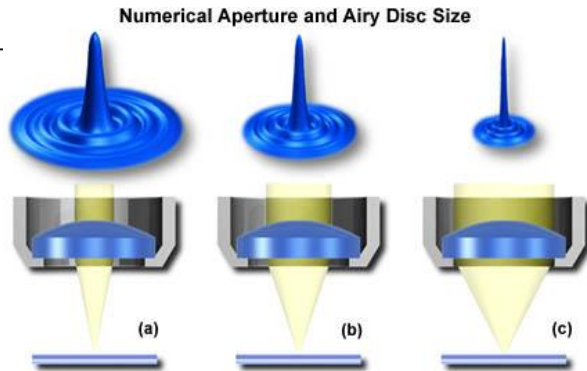
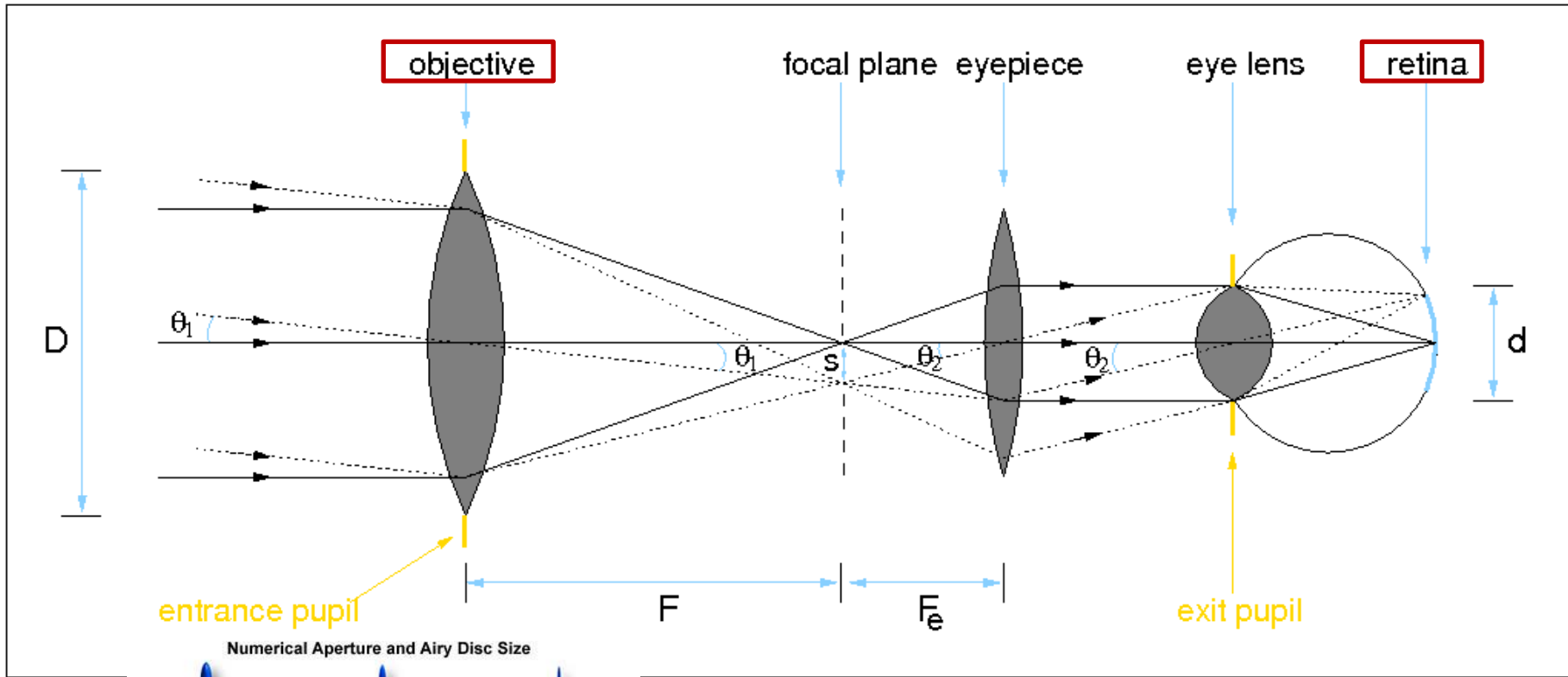


Figure 4

Two distant point sources separated by an angle θ_1 will produce two point images separated by a distance $s = F\theta_1$ in the focal plane of the objective
Magnification $= \theta_2 / \theta_1 = F / F_e$

http://www.vikdhillon.staff.shef.ac.uk/teaching/phy217/telescopes/phy217_tel_visual.html

Example: Resolvability



Calculate the minimal resolvable angle of telescopes with **objective diameter $d = 5.0$ cm** and **$d = 50$ cm**, respectively, for visible light with an average wavelength of 550 nm. The resolvable angle of human's eyes is usually **$1'$ (1 minute)**. Find out the **minimal magnification of the above two telescopes for taking full advantage of their resolvability**.

Consider the minimum angular separation between two resolvable objects,

- $\theta_R = 1.22 \frac{\lambda}{d}$, when **$d = 5.0$ cm**, $\theta_R = 1.22 \times \frac{0.55 \times 10^{-4}}{5.0} = 1.3 \times 10^{-5} \text{ rad}$
when **$d = 50$ cm**, $\theta_R = 1.22 \times \frac{0.55 \times 10^{-4}}{50} = 1.3 \times 10^{-6} \text{ rad}$
- The resolvable angle of human's eyes is **$1' = 2.9 \times 10^{-4} \text{ rad}$** . Therefore, for telescope with $d = 5.0$ cm, its magnification should be at least $\frac{2.9 \times 10^{-4} \text{ rad}}{1.3 \times 10^{-5} \text{ rad}} = 22.5$ times. For telescope with $d = 50$ cm, its magnification should be at least $\frac{2.9 \times 10^{-4} \text{ rad}}{1.3 \times 10^{-6} \text{ rad}} = 225$ times.
- This is in agreement with the fact that telescopes with higher magnification usually need to have larger objective lenses.